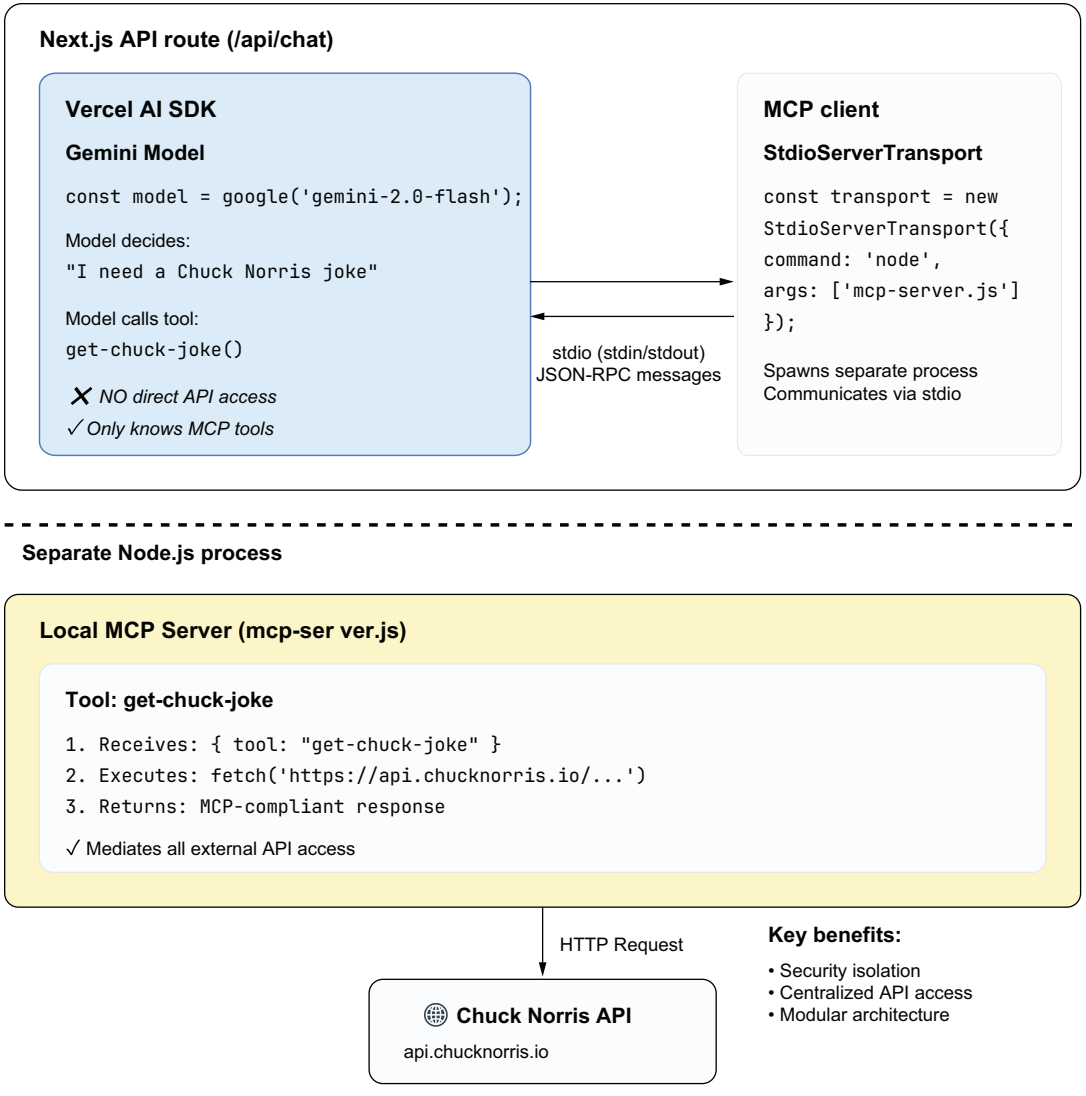


Figure 12.3 How the Vercel AI SDK communicates with a local MCP server through `stdio` transport. The MCP client spawns the server as a separate Node.js process, and they exchange JSON-RPC messages via `stdin`/`stdout`. When the model calls a tool like `get-chuck-joke()`, the request flows through `stdio` to the MCP server, which executes the actual HTTP request and returns the response.

Now start the application from the command line and ask for a few facts about Chuck Norris:

```
$ npm run dev -w ch12/vercel-ai-sdk-mcp
```

Figure 12.4 is a screenshot of the running chat application with MCP tool integration.